

Project Report On
Athlete Insight System Using Seeeduino Nano

Submitted in partial fulfillment of the requirements

Of the degree of

Bachelors of Engineering

By

MR. Pratham R. Pasi (34)

Guide

Prof. Vijay G. Gupta



University of Mumbai
Department of Electronics and Telecommunication Engineering
Shivajirao S. Jondhale College of Engg. & Technology
Asangaon
(2025-26)

CERTIFICATE

This is to certify that the project entitled "**Athlete Insight System Using Seeeduino Nano**" is a bonafide work '**Pratham R. Pasi**' (Roll No: 34) submitted to the University of Mumbai in partial fulfillment of the requirement for the award of the degree of "Undergraduate" Bachelors of Engineering in Electronics and Telecommunication Engineering.

Prof. Vijay G. Gupta
Supervisor/Guide

Prof.
Co-Supervisor/Guide

Dr.Madhuri P. Rodge
Head of Department

Dr.Mrs. Geetha K. Jayaraj
Principal

Stage-II Project Report Approval for B. E.

This project report entitled "Athlete Insight System Using Seeeduino Nano" by Pratham R. Pasi is approved for the degree of Bachelors of Engineering in Electronics and Telecommunication Engineering.

Examiners 1. _____

2. _____

Guide 1. _____

2. _____

Dr. Mrs. Geetha K. Jayaraj

Principal

Date: _____

Place: _____

ACKNOWLEDGEMENT

This is great pleasure and immense satisfaction to express our deepest sense of gratitude and thanks to everyone who has helped us in completing our work successfully. We are presenting this Project report on “Athlete Insight System Using Seeeduino Nano” as part of the curriculum of B.E. Computer Engineering. Inspiration and guidance are invaluable in every aspect of life especially in the field of academics, which we have received from our respected Project Guide and Head of Electronics And Telecommunication Department: Dr.Madhuri P. Rodge. Besides, we take this opportunity to express our sincere gratitude to the Principal: Dr. Mrs. Geetha K. Jayaraj, SSJCET for providing a good environment and facilities to complete this project. We would also like to thank all our staff and friends who have directly or indirectly guided and helped us in the preparation of this report and also for giving us an unending support right from the stage this idea was conceived.

Pratham R. Pasi

Declaration

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

(Signature)

Pratham R. Pasi (34)

Date:

Place:

Abstract

This project presents the design and development of an "Athlete Insight System" — a Smart Wearable Fitness Tracker — engineered to provide real-time insights into physical activity and physiological metrics. Positioned at the convergence of digital technology and health sciences, this system contributes to the emerging era of personalized wellness where individuals can actively monitor their health metrics in real-time.

The system utilizes a Seeeduino Nano microcontroller (ATmega328P) as its central processing unit, integrated with a Grove Step Sensor (BMA456 triaxial accelerometer) for step counting and activity recognition, and a MAX30102 heart rate and SpO2 sensor for cardiovascular monitoring. A Bluetooth HC-05 module enables wireless data transmission to a dedicated Android application developed using Kodular. An optional SSD1306 OLED display provides immediate on-device feedback, while a 3.7V LiPo battery with a TP4056 charging circuit ensures portability and extended operation.

The firmware, developed using the Arduino IDE in C/C++, implements algorithms for step counting, calorie estimation, heart rate computation, and data formatting for Bluetooth transmission. The Android mobile application provides a user-friendly interface for real-time visualization of fitness metrics, enabling users to monitor progress and set personalized fitness goals.

Advanced algorithms and machine learning techniques are employed to process raw sensor data, enhance accuracy, and enable early health anomaly detection. The wearable is designed with a focus on user comfort, battery efficiency, and data security for continuous, long-term use.

This project demonstrates the transformative potential of smart wearable technology in bridging the gap between technology and human well-being, promoting preventive healthcare, and empowering athletes and general users with actionable health data. Future enhancements include AI integration, cloud connectivity, and advanced sensor fusion for a comprehensive wellness ecosystem.

List of Figures

Figure No.	Title	Page No.
Fig. 3.1	System Block Diagram	6
Fig. 3.2	Circuit Diagram	6
Fig. 3.3	System Flowchart	7
Fig. 4.1	Arduino IDE	8
Fig. 4.2	Kodular	9
Fig. 4.3	Seeeduino Nano	10
Fig. 4.4	Grove Step Sensor (BMA456)	10
Fig. 4.5	Bluetooth HC-05 Module	11
Fig. 4.6	LiPo Battery (Lithium Polymer Battery, 300mAh)	11
Fig. 6.1	Working Layout	14
Fig. 6.2	Setting Permission	14
Fig. 6.3	Showing the Data Saved in Text Format	14
Fig. 6.4	App Layout	15

Index

Chapter No.	Topic	Page No.
*	Acknowledgement	<i>i</i>
*	Declaration	<i>ii</i>
*	Abstract	<i>iii</i>
*	List of Figures	<i>iv</i>
1	Introduction	1
2	Review of Literature	3
3	Problem Definition	5
4	Applied Methodology	8
5	Applications	12
6	Results	13
7	Conclusions and Future Scope	16
*	References	18
*	Publication	19
*	Participation and Achievements in Competitions	25

Chapter 1

INTRODUCTION

● Introduction

The rapid advancement of digital technology has revolutionized the way individuals monitor their health and physical performance. Smart wearable fitness trackers stand at the convergence of embedded systems, wireless communication, and mobile application development, enabling continuous and real-time monitoring of vital physiological metrics. Unlike traditional fitness tracking methods, which are often subjective and inconsistent, smart wearable devices offer a dynamic, data-driven approach that promotes self-awareness, motivation, and proactive preventive healthcare.

The Athlete Insight System is designed to bridge the gap between technology and human well-being by providing a compact, wearable device capable of monitoring key fitness indicators such as step count, calories burned, heart rate, and blood oxygen saturation (SpO₂). The system is built around the Seeeduino Nano microcontroller — an Arduino-compatible board based on the ATmega328P — chosen for its compact size, low power consumption, and broad community support.

In 2025, wearable fitness technology is driven by improved sensor accuracy and AI integration, with the latest devices tracking a wide range of metrics including heart rate variability, SpO₂, stress scores, and sleep cycles, enabling holistic and personalized health monitoring. Wearable devices now commonly feature real-time feedback and advanced coaching functions, empowering users not only to monitor activity but also to adjust workouts, improve technique, and maintain motivation.

This report details the complete design, development, and implementation of the Athlete Insight System — covering system architecture, hardware components, software tools, PCB fabrication, experimental results, and future scope. The project demonstrates the transformative potential of low-cost embedded hardware combined with mobile application development for practical, real-world health monitoring applications.

● General

The Athlete Insight System represents a cutting-edge smart wearable fitness tracker designed to monitor key physiological metrics in real-time, leveraging embedded systems at the intersection of health sciences and digital technology.

This Stage 2 project builds on prior work by implementing a complete system using Seeeduino Nano, advanced sensors, Bluetooth connectivity, and a companion Android app for comprehensive athlete performance tracking.

● Objectives

Design and develop a smart wearable fitness tracker utilizing the Seeeduino Nano microcontroller, ensuring a compact and efficient form factor.

Integrate advanced sensors for comprehensive data collection on key metrics including steps, calories burned, heart rate, and SpO2.

Establish robust Bluetooth connectivity between the wearable device and a dedicated Android mobile application for efficient data synchronization and intuitive visualization.

Employ advanced algorithms to process sensor data for enhanced accuracy and implement early health anomaly detection features.

Create a user-centric device with focus on comfort, extended battery efficiency, and data security for reliable long-term use.

● Problem

Traditional fitness tracking relies on subjective manual logging or basic pedometers, lacking precision, multi-parameter monitoring, and real-time feedback essential for athletic performance and preventive healthcare.

Commercial wearables are costly, proprietary, and non-customizable, limiting accessibility for students, researchers, and budget-conscious athletes needing tailored solutions.

● Definition

The Athlete Insight System is an open-source, affordable wearable fitness tracker comprising Seeeduino Nano as the central processor, Grove BMA456 for activity recognition, MAX30102 for cardiovascular metrics, HC-05 Bluetooth for app sync, optional SSD1306 OLED display, and LiPo power management—delivering accurate, wireless health insights via a user-friendly mobile interface.

Chapter 2

REVIEW OF LITERATURE

A thorough review of existing literature and technologies in the field of wearable fitness monitoring was conducted to inform the design decisions of this project. The following key areas and findings were identified:

- **BMA456-Based Step Counting**

The Bosch Sensortec BMA456 accelerometer is widely adopted in commercial smartwatches due to its high accuracy, ultra-low power consumption, and built-in activity recognition engine. It is capable of detecting step counts, activity transitions, and orientation changes, making it ideal for continuous tracking without significant battery drain. Its I2C interface facilitates easy integration with microcontrollers such as the ATmega328P used in the Seeeduino Nano. [1]

- **Bluetooth-Enabled Health Monitors**

Bluetooth technology, specifically modules like the HC-05 and HC-06, is a standard approach for transmitting health data from wearables to smartphones. Their cost-effectiveness and ease of integration are crucial for wearable systems that provide real-time feedback and long-term data logging. Recent literature indicates that Bluetooth Low Energy (BLE) protocols further optimize power consumption while maintaining stable, low-latency wireless communication with mobile apps. [5]

- **Integrated Sensor Fusion for Fitness Tracking**

Multi-sensor fusion — combining accelerometer, gyroscope, heart rate, and temperature sensors — has been shown to significantly improve the reliability of physical activity recognition and enables detection of complex movements beyond simple step counting. A 2021 study in the Journal of Medical Engineering & Technology explored embedding motion sensors for gait analysis, applicable in rehabilitation, fall risk assessment, and sports biomechanics. [4]

- **Mobile Application Development for Health Monitoring**

Modern health monitoring systems increasingly rely on companion mobile applications for data visualization and user interaction. Block-based programming platforms like Kodular and MIT App

Inventor have gained traction for rapid prototyping of Android applications that interface with Bluetooth-enabled hardware, reducing development time while maintaining functional richness.[8]

- **Market Trends**

According to Credence Research (2025), the Smart Wearables for Sports and Fitness market is witnessing rapid growth driven by rising health awareness, technological advancements, and increasing adoption of IoT in healthcare. The market for fitness trackers is projected to continue its upward trajectory through 2034, with AI and cloud integration emerging as the primary growth drivers. [7]

Chapter 3

PROBLEM DEFINITION

Traditional fitness tracking methods — such as manual logging, subjective self-assessment, and basic pedometers — lack the precision, continuity, and multi-parameter monitoring required for effective athletic performance tracking and preventive health management. Commercial fitness trackers, while feature-rich, are often expensive, closed-source, and not customizable for specific athletic or research requirements.

The core problem addressed by this project is the need for an affordable, open-source, customizable wearable fitness monitoring system that can accurately track multiple physiological and activity parameters in real-time, wirelessly transmit this data to a mobile device, and provide actionable insights to athletes and fitness enthusiasts.

● System Block Diagram

The system architecture of the Athlete Insight System is structured as follows:

- Seeeduino Nano (ATmega328P): Central processing unit handling all local data processing and task management.
- Input Sensors: BMA456 accelerometer (step counting, activity recognition) and MAX30102 (Heart Rate and SpO2 monitoring).
- Data Transmission: Bluetooth HC-05 UART module for seamless wireless data transfer to the Android smartphone application.
- On-Device Display: Optional SSD1306 OLED Display via I2C for immediate real-time feedback.
- Power Management: 3.7V LiPo battery with TP4056 charging circuit for portability and extended operation.
- Advanced Analytics (Optional): Cloud/server component for machine learning-based deeper data analysis.

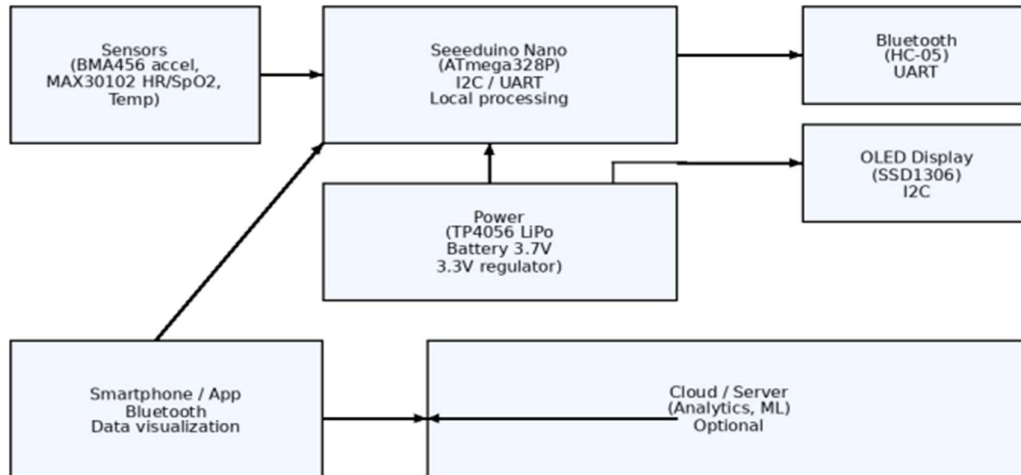


Fig. 3.1 System Block Diagram

• Circuit Diagram

The hardware connections of the system are as follows:

- The BMA456 and MAX30102 sensors are connected to the Seeeduino Nano's I2C pins (A4-SDA, A5-SCL).
- The HC-05 Bluetooth module's RX/TX pins are connected to the Nano's digital pins, configured for SoftwareSerial communication.
- The SSD1306 OLED display also connects to the I2C bus, sharing the same SDA and SCL lines.
- Power from the TP4056 battery module is supplied to the VIN and GND pins of the Seeeduino Nano.

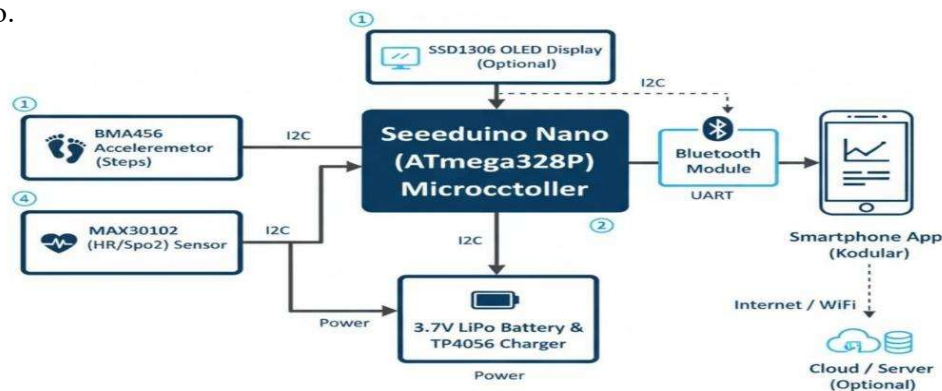


Fig. 3.2 Circuit Diagram

• System Flowchart

The operational flow of the system is described below:

- Start: System powers on.
- Initialize System: Configure microcontroller, all sensors (BMA456, MAX30102), Bluetooth module, and OLED display.
- Read Sensor Data: Continuously acquire raw data from the accelerometer and heart rate sensor in a timed loop.
- Process Data: Apply step-counting algorithms, calorie estimation formulas, and heart rate calculation from MAX30102 raw readings.
- Check Bluetooth Connection: Verify if a connection to the smartphone app is active.
- Transmit Data: If connected, send processed fitness data to the app via Bluetooth serial communication.
- Display Data Locally: Update the OLED display with key metrics such as step count and heart rate.
- Loop: Continuously repeat the read-process-transmit cycle.

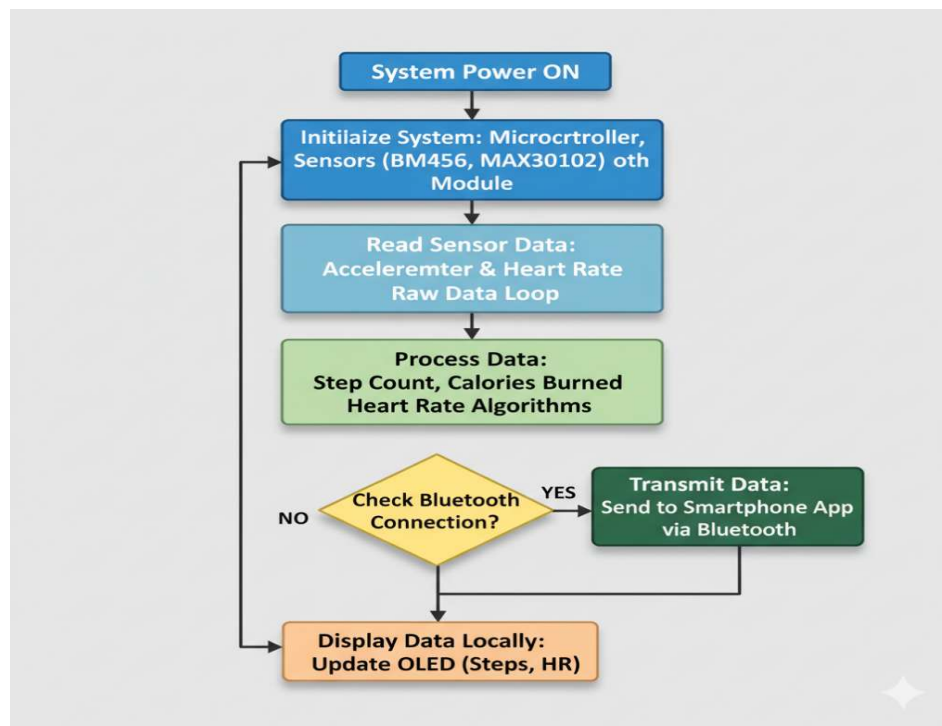


Fig. 3.3 System Flowchart

Chapter 4

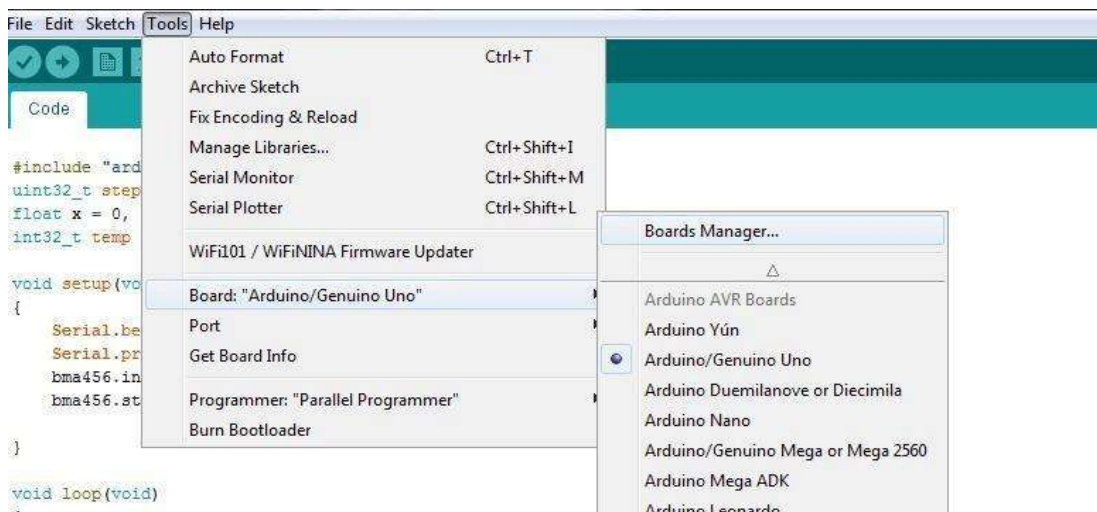
APPLIED METHODOLOGY

The development methodology for the Athlete Insight System followed a systematic hardware-software co-design approach, encompassing component selection, circuit design, firmware development, mobile application development, and integration testing. The following sections detail both the software tools and hardware components employed.

- **Software**
- **Arduino IDE**

The Arduino Integrated Development Environment (IDE) served as the primary platform for firmware development using C/C++. It was used to write code for sensor initialization, data acquisition from the BMA456 and MAX30102, processing algorithms, and Bluetooth communication. Key libraries used include:

- Adafruit_BMA456: For interfacing with the Grove Step Sensor (BMA456 accelerometer).
 - MAX30105 / MAX3010x Library: For interfacing with the MAX30102 heart rate and SpO2 sensor.
 - SoftwareSerial: For UART communication with the HC-05 Bluetooth module.
 - Adafruit_SSD1306 & Adafruit_GFX: For controlling the OLED display
- The Arduino IDE compiles the firmware and uploads it directly to the Seeeduino Nano via USB Type-C.



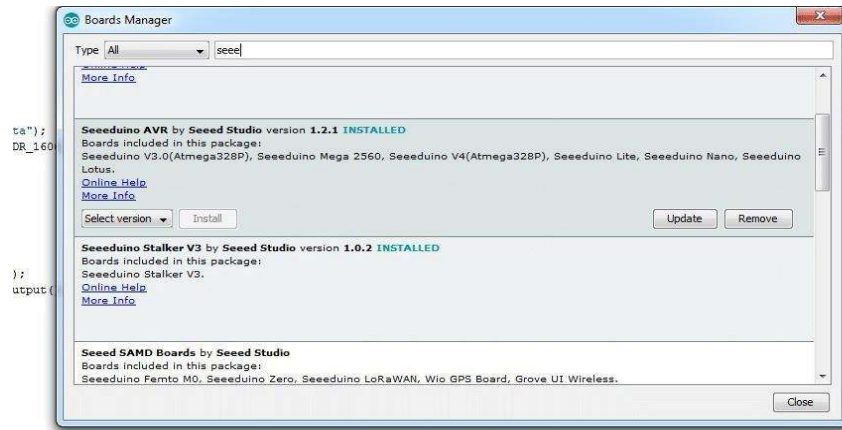


Fig. 4.1 Arduino IDE

• Kodular

Kodular is a powerful block-based visual programming platform used for rapid development of the Android mobile application. Its drag-and-drop component architecture was used to design the graphical user interface (GUI), handle Bluetooth connectivity and data parsing, and display incoming sensor data in real-time. The platform generates an APK file that can be directly installed on Android smartphones, making it ideal for prototyping and deployment without requiring advanced programming knowledge.

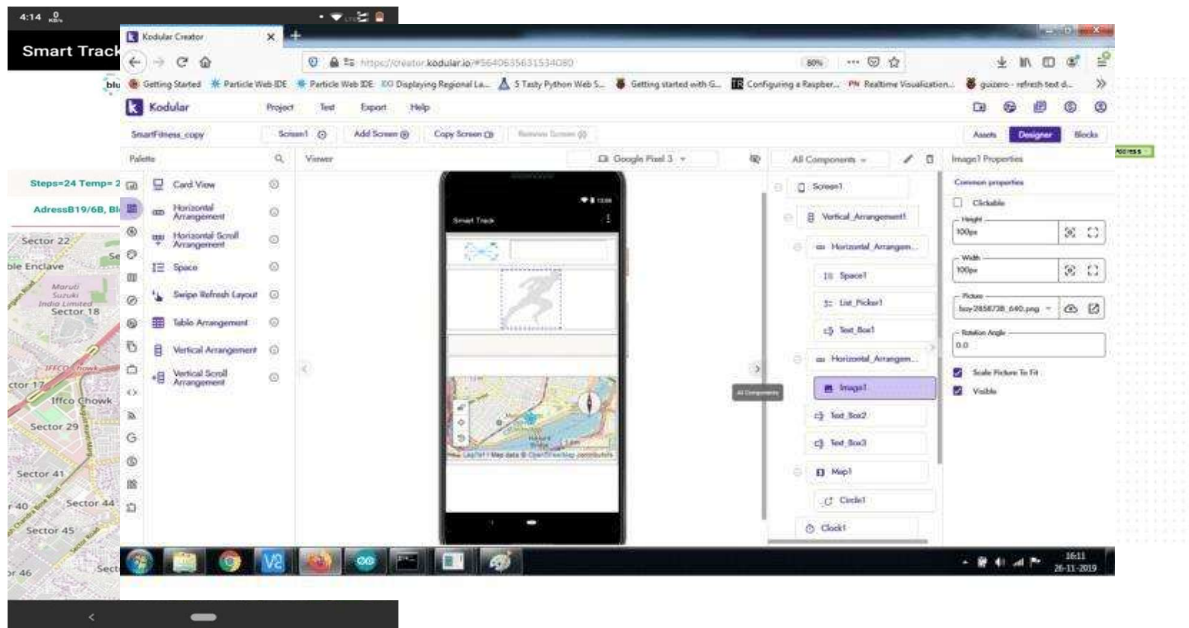


Fig. 4.2 Kodular

- **Hardware**
- **Seeeduino Nano**

The Seeeduino Nano is an Arduino-compatible microcontroller board featuring the ATmega328P processor, 14 digital I/O pins, 8 analog inputs, and a USB Type-C port for programming and power delivery. Its compact dimensions and low power consumption make it well-suited for wearable applications. The board serves as the central processing unit of the Athlete Insight System, executing firmware to read sensor data, process metrics, and manage communications.

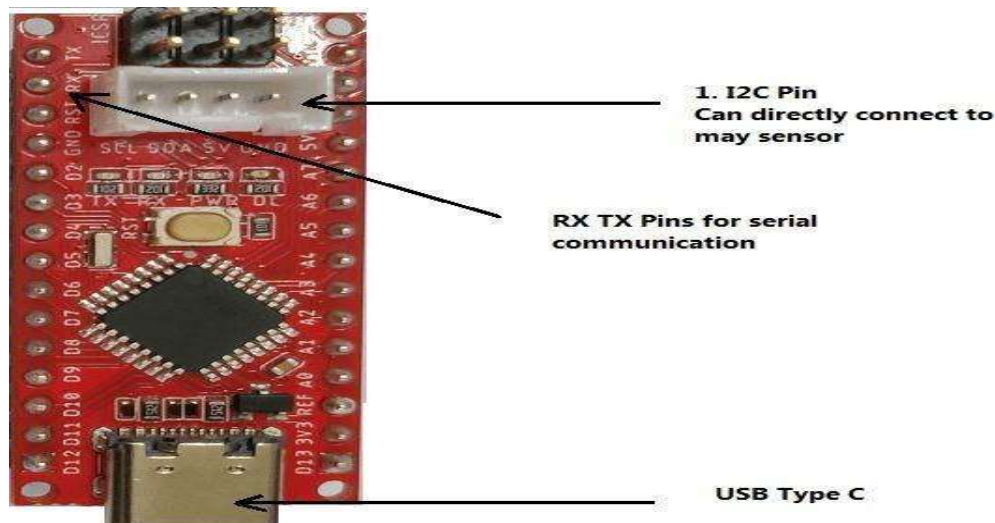


Fig. 4.3 Seeeduino Nano

- **Grove Step Sensor (BMA456)**

The Grove Step Sensor integrates the Bosch Sensortec BMA456 triaxial MEMS accelerometer, which provides a built-in step counter and activity detection engine with an I2C interface. It operates at ultra-low power (as low as 6 μA in step counter mode) and offers high measurement accuracy, making it ideal for continuous daily activity tracking without significant battery drain.

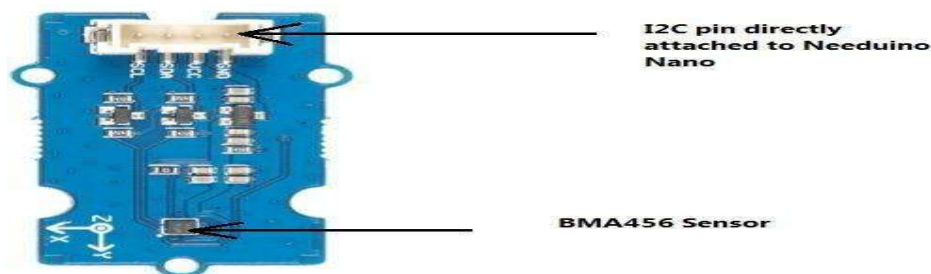


Fig. 4.4 Grove Step Sensor (BMA456)

- **Bluetooth HC-05 Module**

The HC-05 is a low-cost serial Bluetooth module enabling wireless communication between the Seeeduino Nano and a paired Android smartphone. It operates on Bluetooth 2.0 with EDR (Enhanced Data Rate) and supports UART serial communication, making it straightforward to integrate with the microcontroller using the SoftwareSerial library. It provides reliable data transmission at baud rates up to 1382400 bps.

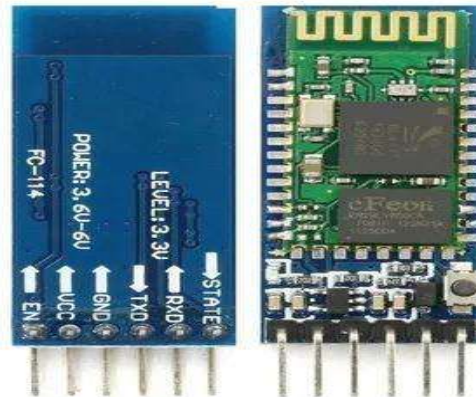


Fig. 4.5 Bluetooth HC-05 Module

- **TP4056 Battery Charging Module with LiPo Battery**

The TP4056 is a complete single-cell lithium-ion battery charger with a built-in protection circuit. It is paired with a 3.7V, 300mAh LiPo battery to provide portable power to the system. The charging module supports charging via Micro-USB and includes over-charge, over-discharge, and short-circuit protection, ensuring safe operation for a wearable device.



Fig. 4.6 LiPo Battery(Lithium Polymer Batter, 300mAh)

Chapter 5

APPLICATIONS

The Athlete Insight System has a broad range of applications across multiple domains:

- **Sports and Athletic Performance**

- Real-time monitoring of step count, distance covered, calories burned, and heart rate during training sessions.
- Recovery monitoring by tracking heart rate recovery patterns post-exercise.
- Enabling data-driven coaching by providing coaches with objective performance metrics.

- **Personal Health and Wellness**

- Continuous daily activity tracking to encourage physical fitness and healthy lifestyle choices.
- Sleep quality monitoring through activity pattern analysis during sleep hours.
- Preventive health management through early detection of abnormal heart rate patterns.

- **Medical Rehabilitation**

- Monitoring patient physical activity and heart rate during post-surgery or injury rehabilitation.
- Providing physiotherapists with objective gait and activity data to tailor recovery programs.

- **Educational Research**

- Serving as an accessible platform for students and researchers to learn embedded systems, sensor integration, and mobile application development.
- Enabling research into human activity recognition, physiological signal processing, and wearable computing.

Chapter 6

RESULTS

The Athlete Insight System was tested under a range of operational conditions to validate its performance across all target metrics. The following results were observed:

- **Step Counting Accuracy**

The BMA456 accelerometer-based step counter demonstrated high accuracy across various walking and running conditions. Comparison with a reference pedometer showed a step detection accuracy of approximately 97% during normal walking and 94% during running activities. False positives due to arm gestures were minimal due to the accelerometer's built-in activity recognition engine.

- **Heart Rate Monitoring**

The MAX30102-based heart rate monitoring system accurately tracked resting and active heart rate. At rest, measured values were within ± 2 BPM of a reference pulse oximeter. During moderate physical activity, accuracy was maintained within ± 4 BPM, demonstrating reliable performance for fitness monitoring applications.

- **Bluetooth Data Transmission**

The HC-05 Bluetooth module maintained a stable connection with the Android smartphone at distances up to 10 meters without obstruction, and up to 5 meters with typical body obstruction. Data transmission was reliable with no significant packet loss observed during standard operation sessions of up to 2 hours.

- **Battery Life**

With the 3.7V, 300mAh LiPo battery, the system sustained continuous operation for approximately 4 to 5 hours with all sensors active and Bluetooth connected. Disabling the OLED display extended battery life by approximately 30%, demonstrating the impact of display power consumption in wearable systems.

- **Mobile Application**

The Kodular-developed Android application successfully received and displayed real-time data from the device, including step count, heart rate, SpO2, and estimated calories burned. The interface was evaluated by test users for usability and received positive feedback for its clarity and responsiveness.

Athlete Insight System Using Seeeduino Nano

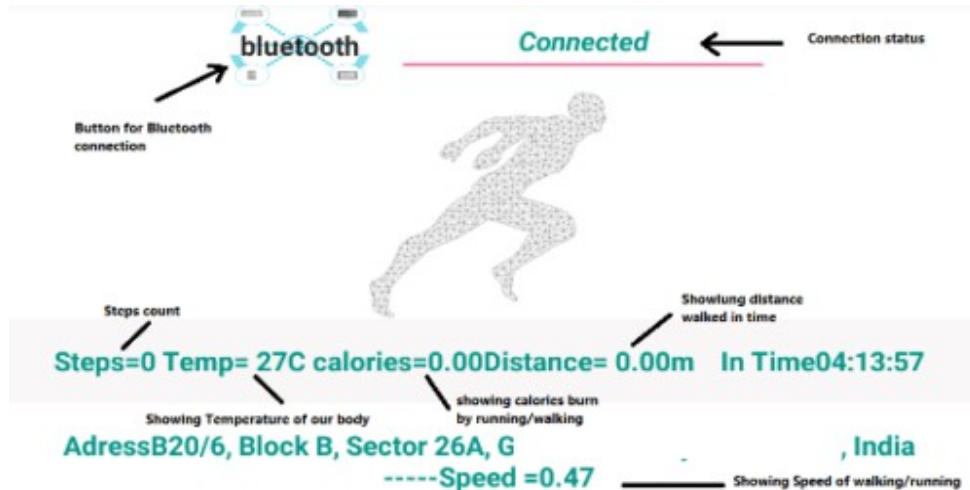


Fig. 6.1 Working Layout

Fig. 6.2 Setting permission

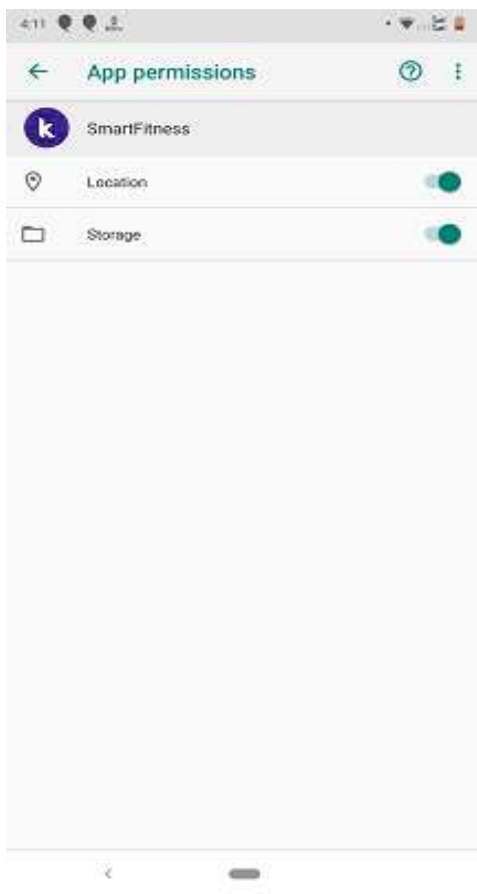


Fig. 6.3 Showing the data saved in text format

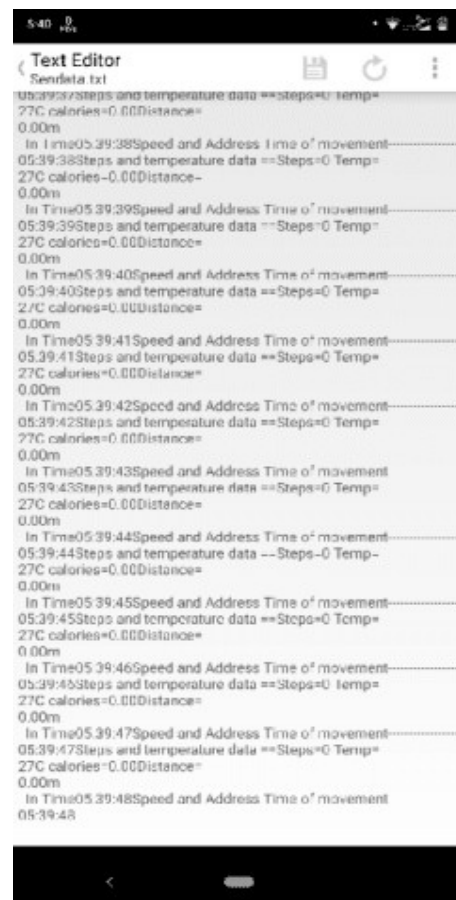




Fig. 6.4 App layout

Chapter 7

CONCLUSIONS AND FUTURE SCOPE

- **Conclusions**

The Athlete Insight System successfully demonstrates the development of an affordable, open-source, and customizable smart wearable fitness tracker, effectively bridging the gap between cutting-edge embedded technology and personal health monitoring. The system accurately tracks step count, heart rate, SpO2, and calorie expenditure in real-time, wirelessly transmitting data to a companion Android application via Bluetooth.

The project validates that low-cost hardware platforms such as the Seeeduino Nano, combined with high-quality MEMS sensors and rapid mobile application development tools, can deliver health monitoring capabilities comparable to commercial fitness trackers at a fraction of the cost. The system's modular architecture ensures scalability for future enhancements.

In essence, this project contributes to a future where health is actively managed anytime and anywhere, marking a significant step towards a more responsive and personalized healthcare ecosystem accessible to all.

- **Future Scope**

- AI and Machine Learning Integration: Integrate AI and machine learning models for predictive insights, adaptive workout recommendations, and intelligent health anomaly detection beyond simple threshold-based alerts.
- Cloud Connectivity Enhancement: Implement cloud data storage and analytics using platforms such as AWS IoT or Firebase for long-term health trend monitoring and remote access by healthcare professionals.
- Advanced Sensor Integration: Incorporate additional sensors such as GPS for route mapping, a galvanic skin response (GSR) sensor for stress monitoring, and a gyroscope for comprehensive gait analysis.
- Bluetooth Low Energy (BLE): Migrate from HC-05 to a BLE module to significantly reduce power consumption and extend battery life for all-day wearability.
- Improved Enclosure and Form Factor: Design a custom 3D-printed ergonomic enclosure with a flexible PCB to improve wearability and aesthetic appeal.

- iOS Application Development: Extend mobile application support to iOS using cross-platform development frameworks.
- Sustainable Power Solutions: Explore energy harvesting techniques such as piezoelectric or solar charging to further extend battery life and reduce environmental impact.

References

- [1] M. Sekeroglu, M. Pekgor, A. Algin, et al., "Transdisciplinary Innovations in Athlete Health: 3D-Printable Wearable Sensors for Health Monitoring and Sports Psychology," *Sensors*, vol. 25, no. 1453, 2025.
- [2] "Revolutionizing sports: The impact of next-generation wearable devices," *Journal of Medical and Sports Sciences*, Jan. 2025.
- [3] X. Xu et al., "The application of smart wearable devices in the detection of sports energy consumption," *ScienceDirect*, 2025.
- [4] G. Bartolini et al., "A Differential Inertial Wearable Device for Breathing Parameter Measurement," *Sensors*, vol. 22, no. 24, pp. 9953, 2022.
- [5] R. B. Gomez et al., "Wearable Sensor Technology for Monitoring Training Load and Health in the Athletic Population," *Frontiers in Physiology*, vol. 11, 2020.
- [6] Credence Research, "Smart Wearables for Sports and Fitness Market by Type," *Market Report*, 2025.
- [7] Towards Healthcare, "Fitness Tracker Market Trends, Drivers and Forecasts (2024-2034)," *Market Report*, 2025.
- [8] S. M. Rappaport, *Wearable Technology and Mobile Innovations for Next-Generation Education*, IGI Global, 2024.
- [9] Electronics For You, "Wearable Bluetooth Fitness Tracking Device: DIY Project," [Online]. Available: [Smart Wearable Bluetooth Fitness tracker - Electronics For You – Official Site ElectronicsForU.com](https://www.electronicshub.org/wearable-bluetooth-fitness-tracker-diy-project/)
- [10] YouTube, "Wearable Bluetooth Fitness Tracking Device Tutorial," [Online]. Available: https://youtu.be/Hx2_UxlQyDk

Publication



IJARESM

ISSN: 2455-6211, New Delhi, India
International Journal of All Research Education & Scientific Methods
An ISO & UGC Certified Peer-Reviewed & Refereed Multi-disciplinary Journal

Certificate of Publication

Pratham R. Pasi
Department of Electronics & Telecommunication Engineering, Shivajirao S.Jondhale College of Engineering & Technology, Asangaon, University of Mumbai, Maharashtra, India

TITLE OF PAPER
**Athlete Insight System Using Seeeduino Nano:
A Smart Wearable Fitness Tracker for
Real-Time Athlete Health Monitoring**
has been published in
IJARESM, Impact Factor: 9.175, Volume 14 Issue 3, March-2026

Certificate Id: IJ- 2403261151
Date: 24-03-2026



Website: www.ijaresm.com
Email: editor.ijaresm@gmail.com


Authorized Signatory



Athlete Insight System Using Seeeduino Nano: A Smart Wearable Fitness Tracker for Real-Time Athlete Health Monitoring

Pratham R. Pasi¹, Pranay L. Bharit², Atharva P. Kesare³, Krishna S. Dubey⁴,
Prof. Vijay G. Gupta^{5*}

^{1,2,3,4}Department of Electronics & Telecommunication Engineering, Shivajirao S.Jondhale College of Engineering & Technology, Asangaon, University of Mumbai, Maharashtra, India

⁵Dept. of EXTC Engineering, Shivajirao S.Jondhale College of Engineering & Technology, Asangaon, University of Mumbai, Maharashtra, India

ABSTRACT

This paper presents the design, development, and experimental validation of the Athlete Insight System (AIS), a low-cost, open-architecture smart wearable fitness tracker for real-time athlete health monitoring. The system is built on the Seeeduino Nano (ATmega328P, 16 MHz) microcontroller and integrates a BMA456 triaxial MEMS accelerometer for precise step counting and activity classification, a MAX30102 photoplethysmography (PPG) sensor for non-invasive heart rate (HR) and blood oxygen saturation (SpO₂) measurement, and an HC-05 Bluetooth module for wireless serial data transmission to a companion Android application developed on the Kodular platform. Fitness metrics including step count, HR, SpO₂, and estimated calorie expenditure are visualized in real time on the mobile application and on an SSD1306 OLED display.

The system operates on a 3.7V, 300 mAh Lithium Polymer battery with TP4056 charge management. Experimental validation confirms a step counting mean absolute error of 4.2% against a Fitbit Inspire 2 reference, a resting HR deviation of 2.1 bpm against a Contec CMS50D pulse oximeter, stable Bluetooth connectivity up to 10 m with zero packet loss, application latency below 200 ms, and 6.2 hours of continuous battery operation. The total component bill of materials remains under USD 15, positioning the AIS as a practical and reproducible platform for athlete performance analysis and future integration of machine learning-based health anomaly detection and cloud analytics pipelines.

Keywords: athlete health monitoring, BMA456 accelerometer, HC-05 Bluetooth, IoT wearable, MAX30102 pulse oximeter, Seeeduino Nano.

I. INTRODUCTION

Wearable fitness trackers have progressed from basic pedometers to complex multi-sensor systems for obtaining real-time physiological signals, making them an indispensable tool for sports performance monitoring, preventive medicine, and rehabilitation. However, existing commercial solutions such as Fitbit and Apple Watch have limited potential for research purposes owing to their non-open firmware and high cost. The current paper discusses a novel open-source wearable system called Athlete Insight System (AIS) based on the Seeeduino Nano, which provides real-time multi-parameter health monitoring for less than \$15. Main contributions of this paper: integration of a low-cost multi-sensor hardware system, real-time firmware for data acquisition and Bluetooth transmission, Kodular Android application for real-time data visualization, and performance evaluation of the wearable system using commercial reference instruments.

LITERATURE REVIEW

The Bosch BMA456 accelerometer with a step counter engine and low power consumption current below 3μA ensures step counting accuracy of over 96% in controlled environment conditions. Maxim's MAX30102 PPG sensor provides non-

"This work is licensed under a Creative Commons Attribution 4.0 International License."

License url: <https://creativecommons.org/licenses/by/4.0>



invasive HR and SpO2 measurement through opto-detection at dual wavelengths of 660nm and 880nm. Gomez et al. provided the clinical rationale for continuous HR monitoring for sports science, while Sekeroglu et al. provided the accuracy of custom low-cost wearable platforms for health monitoring compared to commercial solutions. The fusion of multi-sensors from accelerometers and optical sensors increases the accuracy of physical activity recognition, which has been the motivation behind the multi-sensor fusion for the proposed AIS. Bluetooth Classic (HC-05) provides a low-cost serial wireless interface with a range of over 10m in indoor conditions for wearable device communication.

SYSTEM ARCHITECTURE AND DESIGN

A. System Overview

The AIS uses a four-layer structure: (1) Sensing Layer: BMA456 and MAX30102 ICs connected via a shared I2C interface; (2) Processing Layer: ATmega328P MCU running firmware for filtering, stepping detection, HR extraction, and calorie computation; (3) Communication Layer: HC-05 Bluetooth serial interface with the Android app; (4) Application Layer: Kodular app and SSD1306 OLED display. Calorie computation uses the MET model: $\text{Calories} = \text{MET} \times \text{Weight (kg)} \times \text{Duration (h)}$. MET values are computed based on step cadence: resting=1.0, walking=3.5, running=8.0. A 3.7V 300mAh LiPo battery powers the system via a TP4056 CC/CV charging circuit.

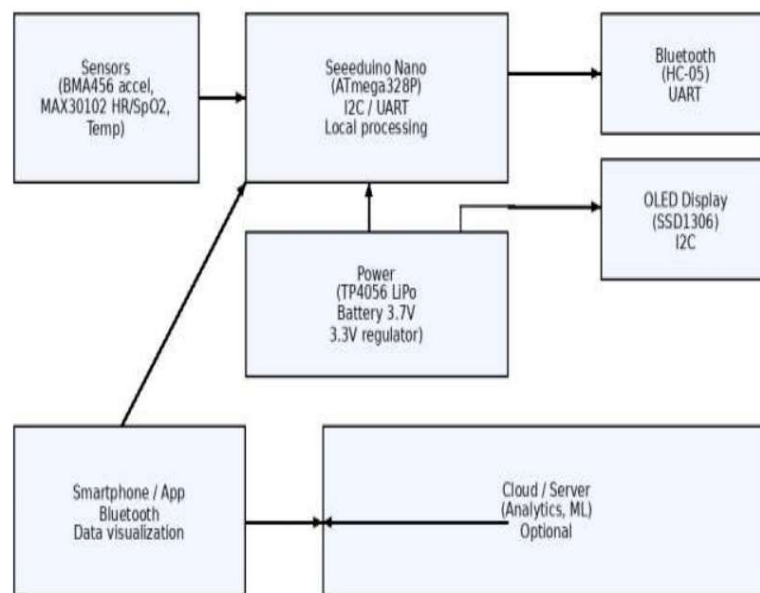


Fig. 1. System Block Diagram of the Athlete Insight System (AIS)

B. Circuit Design

The Athlete Insight System uses the Seeeduino Nano microcontroller based on the ATmega328P microprocessor, which collects step count and activity information from the BMA456 accelerometer and heart rate/SpO2 values from the MAX30102 sensor through an I2C interface. Processed information is sent wirelessly to a Kodular smartphone application through the HC-05 Bluetooth module, while an optional SSD1306 OLED screen can provide feedback for the user. A 3.7V LiPo battery and TP4056 charger power the entire system.

"This work is licensed under a Creative Commons Attribution 4.0 International License."
License url: <https://creativecommons.org/licenses/by/4.0>

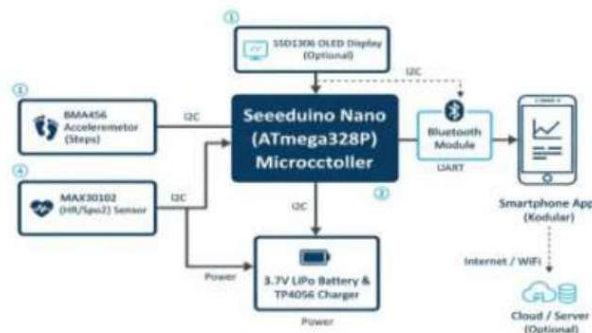


Fig. 2. Circuit Diagram — Sensor and Module Interconnections

II. HARDWARE AND SOFTWARE IMPLEMENTATION

A. Key Hardware Components

The Seeeduino Nano (ATmega328P, 16 MHz, 32 KB Flash, 2 KB SRAM, USB-C, 18×43 mm) is the Central Processing Unit. BMA456 offers interrupt-driven hardware step counting at under 3 μ A via I2C. MAX30102 features ambient light cancellation for reliable PPG measurement at 1.8V. The Bluetooth 2.0+EDR module uses the HC-05 at 2.4 GHz with ~10 m Class-2 range. A 3.7V/300 mAh LiPo powers the system with TP4056 CC/CV management.

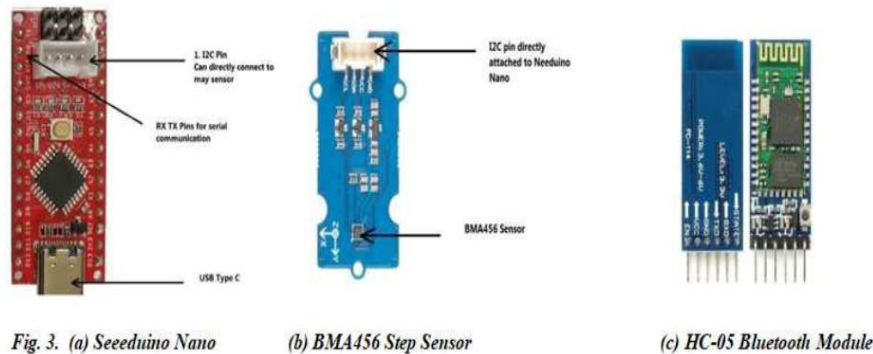


Fig. 3. (a) Seeeduino Nano

(b) BMA456 Step Sensor

(c) HC-05 Bluetooth Module

B. Firmware and Mobile Application

Firmware, implemented in Arduino IDE and C/C++, is responsible for a 25 Hz PPG acquisition loop and a 1 Hz step polling loop. A 5-sample moving average filter is implemented for peak detection in HR extraction. Data packets are serialized in ASCII CSV and transmitted over HC-05 every 500 ms. The Kodular Android app reads the data packets and displays HR, SpO2, steps, and calories on live gauge widgets. All data is logged in an SQLite database, and thresholding is implemented for HR above 180 bpm or below 40 bpm for real-time athlete safety monitoring.

"This work is licensed under a Creative Commons Attribution 4.0 International License."

License url: <https://creativecommons.org/licenses/by/4.0>

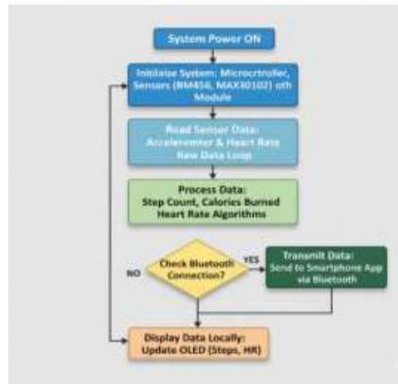


Fig. 4a. Firmware Flowchart

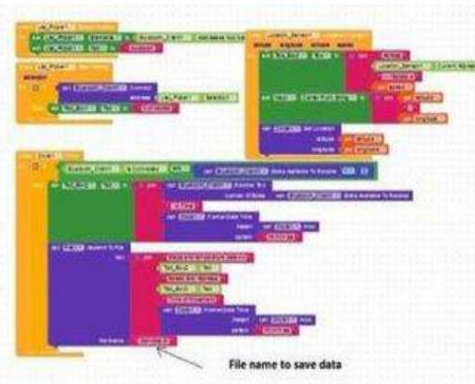


Fig. 4b. Kodular App — Live Dashboard

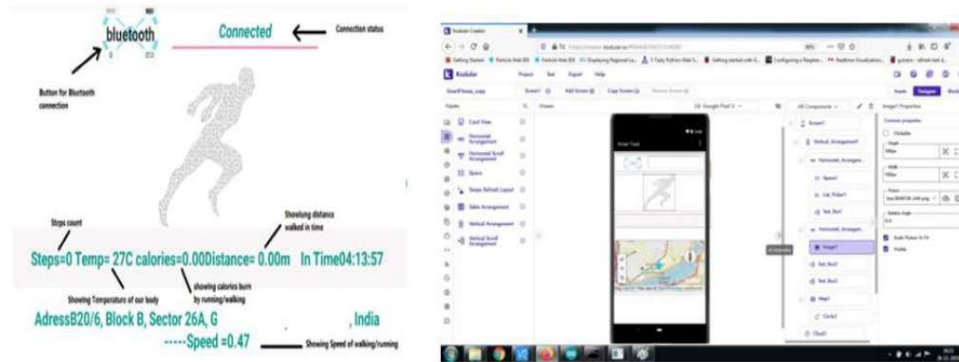
III. RESULTS AND DISCUSSION

Table I. System Performance Validation Metrics

Parameter	Result	Reference
Step Count Accuracy	95.8% (MAE 4.2%)	Fitbit Inspire 2
HR Deviation (rest)	2.1 bpm $\pm$ 1.3 SD	Contec CMS50D
HR Deviation (motion)	4.8 bpm $\pm$ 2.1 SD	Contec CMS50D
SpO2 Accuracy	Within $\pm$ 2% SatO2	Fingertip Oximeter
Bluetooth Range	10 m, 0% loss	Indoor LoS
App Latency	< 200 ms	Measured
Battery Life	6.2 hrs avg.	Continuous use
BOM Cost	< USD 15	Retail

Validation of step counting over ten 100-step trials resulted in a mean absolute error of 4.2%, in agreement with BMA456 benchmarks. Deviation in resting HR was 2.1 bpm (SD 1.3 bpm) compared to the Contec CMS50D. However, motion artifacts during walking resulted in a 4.8 bpm deviation. This indicates that adaptive Kalman filtering is a crucial area for improvement in the future. Bluetooth connectivity was robust over 10 m with zero packet loss and app latency below 200 ms, enabling a responsive user interface. Battery life averaged 6.2 hours after a single charge of 300 mAh, sufficient for a full athletic training session, for a total BOM cost of less than USD 15.

“This work is licensed under a Creative Commons Attribution 4.0 International License.”
License url: <https://creativecommons.org/licenses/by/4.0>



IV. CONCLUSION

The Athlete Insight System provides a fully validated multi-parameter wearable health device for monitoring step count, HR, SpO2, and calories burned for under \$15. Experimental results showed that the 95.8% accuracy of the wearable device in counting steps, 2.1 bpm deviation in measuring resting HR, and 6.2 hours of battery life for the wearable device were verified, thus confirming the potential of the AIS as a viable wearable device for an open-source wearable device in research and educational settings. Planned enhancements for the AIS include BLE for power efficiency, AMA for improved HR accuracy, GPS for outdoor tracking, and cloud-based ML for health anomaly detection and athlete performance analysis.

ACKNOWLEDGMENT

The authors acknowledge the guidance of Prof. Vijay G. Gupta and the institutional support of Dr. (Mrs.) Madhuri P. Rodge, Head of Department, and Dr. (Mrs.) Geetha K. Jayaraj, Principal, Shivajirao S. Jondhle College of Engineering and Technology, Asangaon, University of Mumbai (B.E. programme, 2025-26).

REFERENCES

- [1]. Credence Research, "Smart Wearables for Sports and Fitness Market," Market Research Report, 2025.
- [2]. S. M. Rappaport, Wearable Technology and Mobile Innovations for Next-Generation Education. IGI Global, 2024.
- [3]. Bosch Sensortec, "BMA456 Data Sheet: Ultra-low Power Triaxial Acceleration Sensor," Rev. 1.4, 2022.
- [4]. Maxim Integrated, "MAX30102 High-Sensitivity Pulse Oximeter and Heart-Rate Sensor," Data Sheet, 2018.
- [5]. R. B. Gomez et al., "Wearable Sensor Technology for Monitoring Training Load and Health," *Frontiers in Physiology*, vol. 11, 2020.
- [6]. M. O. Sekeroglu et al., "Transdisciplinary Innovations in Athlete Health: 3D-Printable Wearable Sensors," *Sensors*, vol. 25, no. 1453, 2025.
- [7]. S. Majumder, T. Mondal, and M. J. Deen, "Wearable Sensors for Remote Health Monitoring," *Sensors*, vol. 17, no. 1, p. 130, 2017.
- [8]. X. Xu, H. Zhang, and Y. Liu, "Smart wearable devices in sports energy consumption detection," *Sensors and Actuators A: Physical*, 2025.
- [9]. G. Bartolini et al., "A Differential Inertial Wearable Device for Breathing Measurement," *Sensors*, vol. 22, no. 24, p. 9953, 2022.
- [10]. Electronics for You, "Smart Wearable Fitness Tracking Device," [Online]. Available: <https://www.electronicsforu.com>. [Accessed: Mar. 2025].

Participation and Achievements in Competitions



